

EEE 495 Electrical and Electronics Engineering Design II PROJECT PROPOSAL

Project Title/Product Name: FarmGate Sentinel

Date: 03.10.2025

Team No: 2

Team Members:

Student ID	Name and Surname	e-mail
21802551	Muhammed Gölcük	muhammed.golcuk@ug.bilkent.edu.tr
21903121	Yusuf Semih Gürkök	semih.gurkok@ug.bilkent.edu.tr
22002782	Zeynep İrem İpek	irem.ipek@ug.bilkent.edu.tr
22001648	İsmail Özgenç	ismail.ozgenc@ug.bilkent.edu.tr

1 Statement of the Need / Intended Users

The purpose of this project is to protect animals in farms from predators. Predators (e.g., coyotes, wolves, foxes) around the farm/ranch environments, such as pasture fences, chicken coops, and barns. Off-the-shelf motion lights/cameras often alarm on wind, rain, or machinery, and do not coordinate neighboring properties or store actionable evidence.

This product is needed by small/medium farms and ranches with open paddocks; neighbor co-ops that benefit from pooled alerts, and wildlife managers seeking non-lethal monitoring and deterrence. The means of deterrence are auditory and light-based actuators that are intended to scare the predators.

2 Existing Solutions / Related Work

In previous works in this topic, lights and sounds can deter carnivores at bomas/enclosures, though effects vary and habituation is possible without policy control [1, 2].

In animal intrusion detection ,vision, part, recent farm-surveillance systems combine Internet of Things (IoT) connectivity with modern detectors (e.g.,YOLO 8) for *real-time* animal/intruder detection and cloud notifications [3]. Surveys also catalog animal-intrusion approaches and practical constraints in agricultural settings.

In bioacoustics for howl/bark detection part, automated wolf/canid vocalization detection using spectrogram features and convolutional models is now demonstrated at scale, supporting robust howl detection in the wild [4, 5]. Lightweight environmental sound classifiers optimized for embedded devices further motivate on-node processing [6].

For IoT architecture part, LTE, low-power radios, WiFi with Raspberry Pi gateways and cloud back-ends are widely validated for large-area agriculture telemetry and eventing [7, 8]. These references support our choice of a two-hop design (edge → gateway → cloud).

3 Proposed Work: Approach, Objectives, Functions, and Use Cases

3.1 Approach

The project adopts a **vision-first** strategy. Each node is an ESP32-CAM positioned at a gate or fence line, configured to capture images at a rate of 10 frames per minute. This rate is adjustable by cloud policy, allowing higher sampling (e.g., 30–60 fpm) during high-risk periods or lower rates (2–5 fpm) when risk is minimal. Frames are compressed to JPEG and transmitted via Wi-Fi to a Raspberry Pi gateway, which performs object detection using lightweight convolutional neural networks (YOLO-Nano or MobileNet-SSD). The gateway can also request **burst capture** from a node when motion or suspicious activity is detected, increasing temporal resolution. Audio sensing (howl/bark) is a *secondary objective*, which we plan to implement if we have time. The Raspberry Pi forwards incident thumbnails and metadata to a cloud service over LTE. The cloud aggregates events across multiple properties, reduces false positives, and sends alerts to mobile devices.

3.2 Objectives:

(i) Maintain baseline visual monitoring with adaptive sampling rates. (ii) Achieve reliable detection of animals, persons, and vehicles with low false alarms. (iii) Provide incident evidence (images or clips) with secure storage and retrieval. (iv) Support WAN-based aggregation across properties for regional heatmaps and alerts. (v) Allow automated deterrent with human-in-the-loop activation (spotlight/siren).

3.3 Basic Functions

(i) ESP32-CAM executes a configurable frame schedule, uploading JPEGs to the Raspberry Pi gateway. (ii) The gateway performs detection and event formation, requests bursts if needed, and manages local buffering, activates automated actuator response when configured. (iii) A cloud backend correlates multi-node events, stores evidence, and pushes alerts. (iv) A mobile/web interface displays incident cards with thumbnails and control options. (v) Deterrent actuation (spotlight/siren) is triggered when a confirmed event is captured.

3.4 Representative Use Cases(UC)

UC-A: Baseline patrol. ESP32-CAM captures at 10 fpm; the Raspberry Pi detects an animal, requests burst frames, and uploads a clip. The phone alert contains images and a spotlight control. **UC-B: Bandwidth-aware mode.** When Wi-Fi weakens, the node down-scales to 320×240 and skips near-static frames, but detection continues. **UC-C:**

Audio-assisted escalation(secondary obj.). A detected howl doubles the frame rate for nearby nodes for two minutes, improving recall. **UC-D: Policy update.** The cloud modifies sampling schedules and burst thresholds to adapt to environmental patterns.

4 Technical Description & Initial Design Alternatives

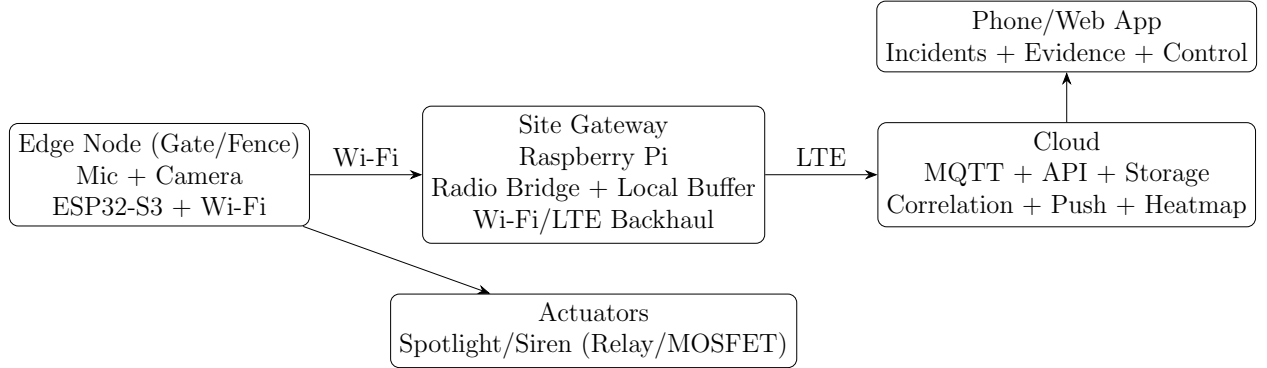


Figure 1: High-level system architecture showing edge nodes, gateway, cloud, deterrents, and mobile app.

4.1 Sensors, compute, communications

Edge node: Digital MEMS microphone, visible-light camera (Raspberry Pi Camera v3), reed switch might be used to detect open or closed gate, and passive infrared sensor and 3-axis accelerometer might be used to ensure both as secondary trust metrics. The ESP32-S3 microcontroller runs MFCC (secondary) + tiny CNN and Wi-Fi.

Gateway: Raspberry Pi 4/5 (Python): radio bridge, local store-and-forward, snapshot/clip verification (lightweight detector), 4.5G Modem or LTE.

Cloud: MQTT broker; Representational State Transfer/WebSocket application programming interface; correlation microservice; object storage for thumbnails/clips; push notifications; incident map/heatmap; role-based access control. Wide Area Network (WAN) is found essential for this project for the following causes: Nodes are installed on different properties and private local networks; only an Internet-reachable cloud can (i) time-align and *fuse* events across sites, (ii) notify roaming owners via mobile data, (iii) preserve evidence & role-based access during local outages, and (iv) learn regional patterns (heatmaps) to cut false positives. A Local Area Network (LAN) alone cannot deliver these outcomes.

4.2 Design alternatives (trade-offs)

(i) *Local radio:* Wi-Fi vs LoRa (longer range, lower bitrate) vs. Bluetooth Low Energy (shorter range, simpler). (ii) *Backhaul:* 4.5G Modem vs LTE 4.5G. (iii) *Verification place:*

Gateway image inference (default) vs. on-node (higher complexity/power). (iv) *Deterrent policy*: Spotlight-only (quiet-hours friendly) vs. spotlight+ siren (higher efficacy).

5 Implementation Plan (Tasks/Roles) & Equipment

Embedded & Audio ML: edge firmware, MFCC + CNN, radio; **Gateway & Cloud**: Pi bridge, MQTT/API, storage, push, correlation; **Hardware/Power**: enclosures, solar, relays, safety; **QA/Field**: datasets, KPI tests, documentation.

Week(s)	Milestones / Tasks	Assigned Member(s)
W1–2	Project Research, Finalizing Ideas	All Members
W3	Project Documentations	Gölcük ,Gürkök
W4	Dataset Research, Model Research, Hardware Research	Gürkök
W5	Raspberry Pi Gateway Server Setup, CNN training	Gölcük, Özgenç
W6	Gateway bridge; cloud skeleton (MQTT, API, push).	İpek,Gölcük
W7	Edge Circuit Design Finish.	İpek, Özgenç
W8	Neighbor fan-out; heatmap; store-and-forward.	İpek, Gürkök
W9	Mobile App Development, Connecting all elements.	Gölcük, Özgenç
W10	WAN demo : snapshot → cloud alert → actuator response.	All Members
W11–13	False-positive reduction; power/latency tuning; robustness.	Gürkok, Özgenç
W14–15	Validation vs KPIs; final report/poster/demo.	All Members

Table 1: Project milestones (15 weeks) with assigned members.

Item	Qty
Raspberry Pi 4/5 (PSU, 32 GB)	1
Wi-Fi Module (gateway)	1
USB LTE 4G/4.5G	1
ESP32–S3 development boards	2
Visible-light cameras (Pi Cam v3 or UVC 1080p or ESP32-CAM)	2
Digital microphones	2
Wi-Fi modules (edge)	2
Reed switches; passive infrared sensors, IMU	2 sets
Spotlight + compact siren + relay/MOSFET driver	1 set
Cabling ,hardware, batteries, custom PCB	–

Table 2: Tentative bill of materials (two nodes + one gateway).

References

- [1] Conservation Evidence, “Use lights and sound to deter predation of livestock by mammals,” evidence synthesis and studies, accessed Sep. 2025. Available: <https://www.conservationevidence.com/actions/2449>
- [2] Conservation Evidence, “Effectiveness of LED lights on bomas in protecting livestock,” accessed Sep. 2025. Available: <https://conservationevidence.com/individual-study/6857>
- [3] T. S. Delwar *et al.*, “Real-Time Farm Surveillance Using Internet of Things and You Only Look Once version 8 for Animal Intrusion Detection,” *Future Internet*, vol. 17, no. 2, 2025. Available: <https://www.mdpi.com/1999-5903/17/2/70>
- [4] N. Makarov *et al.*, “Automated detection of wolf howls using audio classification,” *Scientific Reports*, 2025. Available: <https://www.nature.com/articles/s41598-025-11413-z>
- [5] D. Sossover *et al.*, “Using the BirdNET algorithm to identify wolves, coyotes, and other mammals,” 2023. Available: <https://connormwood.com/wp-content/uploads/2023/11/sossover-et-al-2023-birdnet-for-wolves-and-coyotes.pdf>
- [6] Z. Fang *et al.*, “Fast environmental sound classification based on resource adaptive convolutional neural network,” *Scientific Reports*, 2022. Available: <https://www.nature.com/articles/s41598-022-10382-x>
- [7] G. Codeluppi, L. Cilfone, M. Davoli, and G. Ferrari, “LoRaFarM: A Long-Range Wide-Area Network-Based Smart Farming Modular Internet of Things Architecture,” *Sensors*, vol. 20, no. 7, 2020. Available: <https://www.mdpi.com/1424-8220/20/7/2028>
- [8] M. A. Ahmed *et al.*, “Long Range Based Internet of Things Platform for Remote Monitoring of Large-Scale Agriculture Farms,” *Sensors*, vol. 22, no. 8, 2022. Available: <https://www.mdpi.com/1424-8220/22/8/2824>